

(a) Granular gradation of tea



(b) Grade distribution in bulk tea

Fig. 7: Estimation of manufactured tea grade

ID and pointer to the corresponding image in the image database for future reference.

Granular gradation of tea by e-vision. Instant estimation of manufactured tea grade at drier-mouth is one of the desirable requirements for quality tea production. It is very difficult to find out

the percentage of tea grade at the drier output at any moment of time. An innovative image processing-based solution has been developed to determine the percentage of various tea grades at drier-mouth output as an estimation of consistency in quality tea production (Figs 7(a) and 7(b)).

3 Electronic tongue for taste comparison

Electronic tongue (e-tongue) is an instrument that measures and compares tastes. In a human tongue, chemical compounds responsible for taste are perceived by human taste receptors. In an e-tongue, sensors detect the dissolved compounds. Like human receptors, each sensor has a spectrum of reactions different from the other. The information given by each sensor is complementary and the combination of all sensor results generates a unique fingerprint.

Most of the detection thresholds of sensors are similar to or better than human receptors'. In the biological mechanism,

taste signals are transduced by nerves in the brain into electric signals. E-tongue sensors' process is similar. These generate electric signals as potentiometric variations. Taste-quality perception and recognition is based on building or recognition of activated sensory nerve patterns by the brain on the taste fingerprint of the product. This step is achieved by the e-tongue's statistical software which interprets the sensor data into taste patterns.

The type of taste that is generated is divided into five categories—sour, salty, bitter, sweet and *umami* (delicious). Sourness, which includes hydrochloric acid, acetic acid and citric acid, is created by hydrogen ions; saltiness is registered as sodium chloride; sweetness by sugars; bitterness, which includes chemicals such as quinine and caffeine, is detected through magnesium chloride; and *umami* by monosodium glutamate from seaweed and disodium in meat, fish and mushrooms.

The mechanisms for taste recognition in human and electronic tongues have the same three levels:

1. Receptor level: Taste buds in humans and lipid membrane or novel metal sensors in the e-tongue
2. Circuit level: Neural transmission in humans and transducers in the e-tongue
3. Perceptual level: Cognition in the thalamus in humans and statistical analysis by software in the e-tongue

Receptor level. To detect dissolved organic and inorganic compounds, the sensor-probe assembly is used by the e-tongue at the receptor level. Each probe shows cross-sensitivity and selectivity so that each sensor could concurrently contribute to the detection of most substances found in the liquid matrix. These sensors are composed of an organic coating sensitive to the species to analyse the samples and a transducer that allows

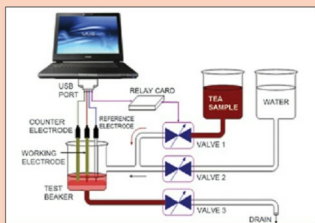


Fig. 8: Internal structure of e-tongue